

***B. Tech Degree VIII Semester (Supplementary) Examination
September 2010***

**EC/EB/EI 804 (C) MECHATRONICS
(2002 Scheme)**

Time : 3 Hours

Maximum Marks : 100

- I. (a) What are the components of a CNC machine? Explain with the help of block diagrams/sketches. (15)
(b) What are the advantages of NC over conventional machine tools? (10)
OR
- II. (a) Explain the terms :
(i) Feed rate (ii) Depth of cut (10)
(iii) Cutting speed (iv) Chips
(b) What are the different types of guide ways used in CNC machines? (15)
- III. (a) Sketch and explain the working of
(i) Fluidic and (15)
(ii) Electronic bistable flip flops.
(b) Explain closed loop and open loop control for CNC machines. (10)
OR
- IV. (a) Explain the working of AC and DC servomotors. (15)
(b) Explain the following feedback devices
(i) Inductosyn
(ii) Resolvers (10)
- V. (a) What is group technology? Explain. (10)
(b) What are the different tape address formats used in NC/CNC programming? Explain each with typical examples. (15)
OR
- VI. (a) What is a “block” in part programming? Write a complete word address and explain each element. (10)
(b) Explain the following terms in programming.
(i) Subroutine
(ii) Mirror image
(iii) Tool compensation (15)
- VII. (a) Explain the different sensors used in robotics, give examples. (15)
(b) Write short notes on :
(i) Lead through programming
(ii) Off line programming (10)
OR
- VIII. (a) Explain
(i) Resolution
(ii) Repeatability of robotic systems (10)
(b) Explain the five basic configurations commonly available in commercial industrial robots. (15)